TauTimp (Onduscular Theory) — Spherical Wave Equation, Muon Neutrino Flux, Cosmological Redshift, and Wheeler's Delayed-Choice Experiment Viorel-Mihai Popescu 21 August 2026

1 TauTimp — Onduscular Theory

1.1 Spherical Wave Equation, Muon Neutrino Flux, Cosmological Redshift, and Wheeler's Delayed-Choice Experiment

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1.2 Abstract

TauTimp (the Onduscular Theory, *Teoria Ondusculară*) proposes a single variable-coefficient spherical wave equation as the generator of both cosmological structure and quantum-measurement behaviour. This document unifies three working notes into one coherent presentation:

1. **Field theory and muon neutrino flux** — the onduscular wave equation is separated into a temporal harmonic equation and a radial equation that reduces to the Airy equation, giving a discrete eigenvalue spectrum λ_n fixed by the boundary condition at the onduscular sphere radius $2R_s$. The associated muon neutrino flux $P(r,t)$ is obtained in closed form.
2. **Cosmological redshift** — the same flux, expressed as a spatial flux growth rate $\Lambda(r)$, is shown to reproduce the observed comoving size of the Local Universe (≈ 93 Gly) and to yield the Hubble constant $H_0 = 67.41$ km/s/Mpc as a *postdiction* of the onduscular geometry rather than a free parameter, without invoking recessional velocity or metric expansion.
3. **Wheeler's delayed-choice experiment** — the single-atom realization of Wheeler's gedanken experiment (Truscott, Manning, Khakimov & Dall) is reinterpreted through Cramer's Transactional Interpretation of Quantum Mechanics (TIQM), with the TauTimp-specific assignment of two propagation speeds: the retarded **offer wave** (the directly observable photon or atom) travels at c, while the advanced **confirmation wave** — identified with the longitudinal entanglement particle — travels at $V = c \times 10^9$.

Together, these three strands describe TauTimp as a framework in which the same onduscular medium — sourced by muon-neutrino propagation and entanglement-mediated confirmation — governs large-scale cosmic expansion and small-scale quantum non-locality.

1.3 0. Physical Parameters

Symbol	Value	Units	Notes
R_s	8.3855194×10^{25}	m	Onduscular sphere radius
R_s	8.86369	Gly	= 8.86369 billion light-years
$2R_s$	1.6771×10^{26}	m	= 17.727 Gly (onduscular event horizon)
R	4.2355×10^{22}	m	= 4.477 million light-years (torsion boundary, $2R$)
$r_{\min}$ (cosmological)	0.431	Gly	Epoch of first star formation
$r_{\min}$ (muon flux)	0.290	Gly	Inner onduscular boundary used for the H_0 derivation
c	2.99792458×10^8	m/s	Speed of light
V	$c \times 10^9 = 2.998 \times 10^{17}$	m/s	Superluminal entanglement / confirmation-wave speed (§14–§15)
H_0	67.41	km/s/Mpc	Planck 2018
c/H_0	1.37229×10^{26}	m = 14.505 Gly	Hubble horizon R_H

2 Part I — Field Theory: Spherical Wave Equation and Muon Neutrino Flux

2.1 1. Governing Equation — Onduscular Wave Equation

2.1.1 1.1 The Variable-Coefficient Spherical Wave Equation

The onduscular wave equation for a scalar field $H(r,\theta,\phi,t)$ is $\frac{R_s}{r}\,\Delta H(r,\theta,t) = \frac{1}{c^2}\frac{\partial^2 H}{\partial t^2}$

Under **spherical symmetry** ($H = f(r,t)$, no θ,ϕ dependence), the spherical Laplacian reduces to $\Delta f = f_{rr} + 2/r\, f_r$, giving $\boxed{\frac{R_s}{r}\left(f_{rr} + \frac{2}{r} f_r\right) = \frac{1}{c^2}\, f_{tt}}$

where R_s is the onduscular cosmological radius, r the radial coordinate, t cosmic time, and c the speed of light. The variable coefficient R_s/r, absent from the standard wave equation, encodes the coupling between the onduscular medium's geometry and wave propagation.

2.1.2 1.2 Physical Interpretation of the Variable Coefficient

The factor R_s/r acts as a **geometric coupling**: at small $r \ll R_s$ the coupling is large (strong onduscular influence); at the boundary $r = 2R_s$ it diminishes to $1/2$. This spatial gradient of coupling is the fundamental origin of the non-linear spatial flux profile $\Lambda(r)$ derived in Part II.

2.1.3 1.3 Separation of Variables

Writing $f(r,t) = g(r)\,u(t)$ and substituting into (1):

$$\frac{R_s}{r\,g}\left(g'' + \frac{2}{r}g'\right) = \frac{u''}{c^2\,u} = -\lambda^2$$

which separates into a temporal equation and a radial equation:

$$u'' + \lambda^2 c^2 u = 0$$

$$g'' + \frac{2}{r}g' + \frac{\lambda^2 r}{R_s}g = 0$$

Time equation — exact solution. Equation (T) is simple harmonic: $\boxed{u(t) = A\cos(\lambda c\,t) + B\sin(\lambda c\,t)}$, $\omega = \lambda c$

2.1.4 1.4 Radial Equation — Reduction to the Airy Equation

Substituting $g(r) = r^{-1/2}\phi(r)$ (motivated by the Bessel-type structure of (R)) gives, after multiplying by $r^{1/2}$,

$$\phi'' + \left(\frac{\lambda^2 r}{R_s} - \frac{1}{4r^2}\right)\phi = 0$$

For $r \gg \lambda^{-1}$ (the wave-dominated regime), the $1/(4r^2)$ term is negligible. Defining

$$\xi = \left(\frac{\lambda^2}{R_s}\right)^{1/3} r = \alpha\, r, \qquad \alpha = \left(\frac{\lambda^2}{R_s}\right)^{1/3}$$

reduces the radial equation to the **Airy equation**: $\boxed{\frac{d^2\phi}{d\xi^2} = \xi\,\phi}$

with general solution $\phi(\xi) = C_1\,\mathrm{Ai}(\xi) + C_2\,\mathrm{Bi}(\xi)$.

2.1.5 1.5 Boundary Conditions and Quantization

The boundary condition $f(2R_s, 0) = 0$ requires $g(2R_s) = 0$, i.e. $\phi(\xi_n) = 0$ at $\xi_n = \alpha \cdot 2R_s$. Physical (bounded-at-infinity) solutions require $C_2 = 0$ (since Bi diverges), leaving $\phi(\xi) = C_1\,\mathrm{Ai}(\xi)$. Requiring $\mathrm{Ai}(\alpha_n \cdot 2R_s) = 0$ at the n-th zero a_n of the Airy function fixes the eigenvalue spectrum:

$$\alpha_n \cdot 2R_s = -a_n \quad \Rightarrow \quad \boxed{\lambda_n = \frac{(-a_n)^{3/2}}{2\sqrt{R_s}}}$$

The first Airy zeros are $a_1 \approx -2.3381$, $a_2 \approx -4.0879$, $a_3 \approx -5.5206$.

2.1.6 1.6 Radial Eigensolutions and the Full Field

$$g_n(r) = r^{-1/2}\,\mathrm{Ai}(\alpha_n r), \qquad \alpha_n = \left(\frac{\lambda_n^2}{R_s}\right)^{1/3}$$

$$\boxed{f(r,t) = \sum_{n=1}^{\infty} r^{-1/2}\,\mathrm{Ai}(\alpha_n r)[A_n\cos(\omega_n t) + B_n\sin(\omega_n t)]}, \qquad \omega_n = \lambda_n c$$

with A_n, B_n fixed by the initial conditions $f(r,0)$, $f_t(r,0)$.

For the general non-symmetric field, expand in real spherical harmonics: $H(r,\theta,\phi,t) = \sum_{\ell,m} f_{\ell m}(r,t)\,Y_\ell^m(\theta,\phi)$, with each radial mode obeying $g_\ell'' + 2/r\,g_\ell' + (\lambda^2 r/R_s - \ell(\ell+1)/r^2)g_\ell = 0$; $\ell = 0$ recovers §1.1–1.6 exactly.

2.2 2. Muon Neutrino Speed Constraint

The muon neutrino (μ_ν) is treated as a Dirac neutrino with spin quantum number $s = -1/2$, travelling at c. The condition $|\boldsymbol{v}_\mu| = c$ is expressed via the total derivative of the onduscular field along the neutrino's path $(r,\ ct)$:

$$\frac{df(r,t)}{dt} = \frac{\partial[u(t)g(r)]}{\partial t} + \frac{\partial[u(t)g(r)]}{\partial r}\cdot\frac{dr}{dt} = c \qquad (2)$$

Since $dr/dt = c$ at every r (the muon neutrino propagates at c), (2) becomes

$$g(r)\,\frac{\partial u}{\partial t} + u(t)\,\frac{\partial g}{\partial r}\cdot c = c \qquad (3)$$

This is the **space-growth constraint**: the total rate of change of the onduscular field equals c at all times and positions — the space-generation rate is constant and equal to c.

2.3 3. Muon Neutrino Flux Equation

2.3.1 3.1 Wave Equation for the Flux Density

The muon neutrino flux $P(r,t)$ (probability/particle flux per unit radial interval) satisfies the standard 1-D wave equation in the radial coordinate:

$$\frac{1}{c^2}\frac{\partial^2 P}{\partial t^2} - \frac{\partial^2 P}{\partial r^2} = 0$$

Boundary condition: $P(2R_s, t) = 0$ (zero flux at the onduscular boundary). **Initial derivative condition:** $\partial_r P|_{r=0,t=0} = -R = -4.2355 \times 10^{22}$ m (fixed by the onduscular parameter R). **Normalization:** $\int_{r_{\min}}^{2R_s} P\,(r,0)\,dr = 1$.

2.3.2 3.2 Degree-2 Polynomial Solution at $t = 0$

A degree-2 polynomial $P_0(r) = p_2 r^2 + p_1 r + p_0$ satisfies (4) at $t = 0$ for any coefficients (since $P_{rr} = 2p_2 = \text{const}$). The bell-shaped, physically natural, boundary-vanishing choice is the **factored form**:

$$\boxed{P(r,0) = C\,(r - r_{\min})(2R_s - r)}$$

which vanishes at the inner boundary $r = r_{\min}$ and outer boundary $r = 2R_s$, and is positive on $(r_{\min}, 2R_s)$.

2.3.3 3.3 General Time Evolution — d'Alembert Solution

The general solution of the wave equation (4) is $P(r,t) = F(r + ct) + G(r - ct)$, determined by the initial data $P(r,0)$ and $P_1(r) = \partial_t P|_{t=0}$. For $P_1 = 0$:

$$\boxed{P(r,t) = C[(r - r_{\min})(2R_s - r) - c^2 t^2]}$$

which at $t = 0$ reduces to $P(r,0) = C(r - r_{\min})(2R_s - r)$.

2.4 4. Muon Neutrino Flux Density $F(r)$ / $\Lambda_{\text{flux}}(r)$

2.4.1 4.1 Definition

$$\boxed{\Lambda_{\text{flux}}(r) = \frac{P(r,0)}{4\pi r^2} = \frac{C}{4\pi} \cdot \frac{(r - r_{\min})(2R_s - r)}{r^2}}$$

2.4.2 4.2 Physical Justification — Hubble-Horizon Normalization

Integrating the flux over the radial shell from $r_{\min}$ to the onduscular boundary $2R_s$ and requiring it to equal the Hubble radius $R_H = c/H_0$: $\int_{r_{\min}}^{2R_s} \Lambda_{\text{flux}}(r)\, dr = \frac{c}{H_0}$

Physical interpretation. The flux-integrated column depth equals the Hubble horizon scale $R_H = c/H_0$. The onduscular boundary $r = 2R_s$ marks the edge of the onduscular sphere; the Hubble law $v = H_0 r$ at $r = c/H_0$ marks where recession velocity equals c. This identity links the local neutrino flux distribution to the cosmic expansion rate — a core TauTimp prediction, consistent with $2R_s \approx 17.73$ Gly $\approx 1.222\, R_H$ (the onduscular boundary lies just beyond the Hubble horizon).

2.4.3 4.3 Analytic Evaluation of the Integral

$$\mathcal{G} \equiv \int_{r_{\min}}^{2R_s} \frac{(r - r_{\min})(2R_s - r)}{r^2}\, dr = (r_{\min} + 2R_s)\ln\frac{2R_s}{r_{\min}} - 2(2R_s - r_{\min})$$

Numerically, with $r_{\min} = 0.290$ Gly, $\mathcal{G} = 3.7113 \times 10^{26}$ m.

2.4.4 4.4 Determination of the Normalization Coefficient C

From (7): $\frac{C}{4\pi} \cdot \mathcal{G} = \frac{c}{H_0}$, so $\boxed{C = \frac{4\pi c}{H_0\,\mathcal{G}}} = 4.6465\ \text{m}^{-2}$

2.4.5 4.5 Expanded Polynomial Coefficients

$$P(r,0) = -C\,r^2 + C(r_{\min} + 2R_s)\,r - C\,r_{\min} \cdot 2R_s$$

Coefficient	Expression	Value (SI)
p_2	$-C$	$-4.6465\ \text{m}^{-2}$
p_1	$C(r_{\min} + 2R_s)$	$7.9202 \times 10^{26}\ \text{m}^{-1}$
p_0	$-C\,r_{\min} \cdot 2R_s$	-2.1381×10^{51}

2.4.6 4.6 Hubble Constant from the Integral — Analytic Formula

$$\boxed{H_0 = \frac{4\pi c}{C\left[(r_{\min} + 2R_s)\ln\frac{2R_s}{r_{\min}} - 2(2R_s - r_{\min})\right]}}$$

With $r_{\min} = 0.290$ Gly and $R_s = 8.8637$ Gly: $H_0 = 67.41$ km/s/Mpc, in excellent agreement with Planck 2018 (67.4 ± 0.5 km/s/Mpc). This is a **postdiction** of TauTimp — the onduscular parameters R_s and $r_{\min}$ fix the Hubble constant directly from field-theoretic quantities, without an assumed metric expansion.

3 Part II — Cosmological Redshift from Spatial Flux Growth

3.1 5. Spatial Flux Growth Rate $\Lambda(r)$

3.1.1 5.1 Expanded Form

Solving the PDE (1) and extracting the spatial-growth integrand as a Laurent series about $r = 0$ gives the **spatial flux growth rate**: $\boxed{\Lambda(r) = \frac{43.1072}{r^2} - \frac{0.6533}{r} - 0.12012}$

where r is in billions of light-years (Gly), obtained with $R_s = 8.2$ Gly and $c = 3$ in normalized units, via

$$\Lambda_{\text{flux}}(r)|_{t=0} = \frac{4R_s^2 c^2 - 4c_2 R_s^2}{8\pi c^2} \cdot \frac{1}{r^2} - \frac{R_s}{4\pi} \cdot \frac{1}{r} + \frac{c_2}{8\pi c^2}, \qquad c_2 = -\frac{8c^2 R_s}{7} + \frac{4}{7}c^2$$

3.1.2 5.2 Indefinite Integral

$$\int \Lambda(r)\, dr = -\frac{43.10796}{r} - 0.6532788\ln r - 0.1201207\, r + \text{const}$$

3.2 6. Comoving Distance Prediction

Integrating $\Lambda(r)$ from the epoch of first star formation ($r_{\min} = 0.431$ Gly, 431 Myr after the Big Bang) to the onduscular boundary ($2R_s \approx 17.727$ Gly):

$$\int_{0.431}^{17.727} \Lambda\,(r)\,dr = 93 \text{ Gly}$$

This **reproduces the observed comoving diameter of the Local Universe** (≈ 93 Gly). The lower limit $r = 0.431$ Gly corresponds to the moment space (via μ-neutrino propagation) and time (via τ-neutrino/photon emission from first stars) were simultaneously generated within the Local Universe; the upper limit $2R_s$ is the onduscular event horizon.

Parameter	Value	Physical meaning
$2R_s$	≈ 17.727 Gly	Onduscular event horizon (Local Universe boundary)
$r_{\min}$	0.431 Gly	Epoch of first star formation
$\int \Lambda\, dr$	93 Gly	Total comoving size of Local Universe
H_0	67.41 km/s/Mpc	Derived from onduscular parameters

3.3 7. Physical Interpretation of the Redshift

3.3.1 7.1 TauTimp Mechanism

In ΛCDM, redshift arises from expansion of space encoded in the scale factor $a(t)$. In TauTimp, redshift arises instead from the **spatial flux growth rate** $\Lambda(r)$:

> A photon emitted at comoving distance r travels through an onduscular medium whose metric is set by $\Lambda(r)$. The greater the distance, the more "new space" — generated by μ-neutrino propagation — the photon must traverse, stretching its wavelength.

The standard relation $1 + z = \lambda_{\text{obs}}/\lambda_{\text{rest}}$ is reinterpreted as

$$1 + z(r) = 1 + \frac{\Delta\lambda}{\lambda} \propto \int_0^r \Lambda\,(r')\,dr'$$

3.3.2 7.2 Non-linear Radial Dependence

Because $\Lambda(r) \propto r^{-2}$ at small r, the growth rate is strongly non-linear:

- $r < 4$ **Gly:** $\Lambda(r)$ large; photons redshift steeply per unit comoving distance — steeper-than-Hubble relation at small distances.
- $r \sim 4$–13 **Gly:** $\Lambda(r)$ near zero; space growth per additional Gly negligible — this naturally accounts for the apparent acceleration seen in Type Ia supernova surveys. In TauTimp it is not acceleration but the *decrease of near-zone spatial flux* that creates the illusion.
- $r \to 2R_s$**:** $\Lambda(r) \to 0$; no new space generated at the boundary. Sources at the Local Universe edge carry the maximum accumulated redshift.

3.3.3 7.3 The 5th Force — Torsion

TauTimp introduces a **torsion force** as a 5th fundamental interaction:

- For $r < 2R = 2 \times 4.2355 \times 10^{22}$ m = 8.955 million ly: gravity is **attractive**.
- For $r > 2R$: torsion is **repulsive**, driving the apparent large-scale acceleration.

This explains the filamentary structure of the cosmic web (SDSS observations): stars within galaxy filaments, closer than $2R$, are held by gravity; on scales beyond $2R$, torsion generates the observed voids and web topology.

3.4 8. Galaxy GN-z11 as a Test Case

For GN-z11 ($z \approx 10.96$), TauTimp gives: proper distance at emission ≈ 1.187 Gly; space expansion factor ≈ 12.1 ($1 + z \approx 12.1$); current comoving distance ≈ 14.36 Gly; total light travel time ≈ 13.4 Gyr. A ΛCDM-style cosmology calculator ($\Omega_M = 0.018$, $\Omega_\Lambda = 1$, $H_0 = 69.6$ km/s/Mpc) gives comparable order-of-magnitude values (age at z: 1.39 Gyr; comoving distance 82–84.8 Gly; light travel time ~ 24 Gly), consistent within the framework's parameter space.

3.5 9. Onduscular Hubble Constant Derivation

$$\boxed{H_0 = \frac{4\pi c}{C\left[(r_{\min} + 2R_s)\ln\frac{2R_s}{r_{\min}} - 2(2R_s - r_{\min})\right]}}$$

giving $H_0 = 67.41$ km/s/Mpc, in agreement with Planck 2018 — a **postdiction**: the onduscular parameters fix the Hubble constant analytically.

3.6 10. Summary of Key Results (Parts I–II)

Result	Formula	Value
Onduscular wave equation	$(R_s/r)(f_{rr} + 2f_r/r) = f_{tt}/c^2$	—
Eigenvalue spectrum	$\lambda_n = (-a_n)^{3/2}/(2\sqrt{R_s})$	discrete (Airy zeros)
Muon neutrino flux $P(r, 0)$	$C(r - r_{\min})(2R_s - r)$	—
Normalization C	$4\pi c/(H_0 \mathcal{G})$	4.6465 m^{-2}
Analytic $\mathcal{G}$	$(r_{\min} + 2R_s)\ln(2R_s/r_{\min}) - 2(2R_s - r_{\min})$	3.7113×10^{26} m
Spatial growth rate $\Lambda(r)$	$43.1072/r^2 - 0.6533/r - 0.12012$	Gly units

Result	Formula	Value
Comoving Universe size	$\int_{0.431}^{17.727} \Lambda \, dr$	93 Gly
H_0 (derived)	$4\pi c/(C\mathcal{G})$	67.41 km/s/Mpc
Onduscular boundary	$2R_s$	17.727 Gly
Torsion force boundary	$2R$	8.955×10^6 ly
GN-z11 comoving distance	—	≈ 82–84.8 Gly

4 Part III — Wheeler's Delayed-Choice Experiment and the Transactional Interpretation

Primary sources cited in the original note: - A. G. Manning, R. I. Khakimov, R. G. Dall, A. G. Truscott — *Wheeler's delayed-choice gedanken experiment with a single atom*, Nature Physics (local reference nphys3343.pdf). - J. G. Cramer — *The Transactional Interpretation of Quantum Mechanics and Quantum Nonlocality*, arXiv:1503.00039.

4.1 11. Wave–Particle Duality and the Role of Measurement

Wave-like and particle-like behaviour are both natural manifestations of light and matter; which of these apparently incompatible behaviours is observed is decided by the choice of measuring apparatus — a defining example of the counter-intuitive character of quantum mechanics.

Double-slit experiment. Photons interfere because they are temporarily ionized by tau-neutrino–type particles. In this picture, temporarily ionized photons can pass through the slits one at a time, as a wave, yet still build up the interference pattern.

Mach–Zehnder interferometer. Temporarily ionized photons travel, one at a time, along two paths — 50% on one path, 50% on the other. If the time separation between successive photons is smaller than a quantum of time, they interfere with one another.

4.2 12. Temporal Ionization and Neutrino Energy Thresholds

Almost all particles carry spin (spin angular momentum — $\hbar/2$ for the electron) and are entangled through a mediating particle — a **longitudinal wave** — that propagates faster than light. In this formulation, the propagation speed is fixed at $\boxed{V = c \times 10^9 = 2.998 \times 10^{17} \text{ m/s}}$, and $V = c \times 10^9$ is retained here as the single, consistent value used throughout the TauTimp corpus (see also TauTimp_Gamma_Wave2.md, the source of this value for the gamma field). The wavefunction collapses upon measurement with photons from the optical window 400–700 nm, energy 2–4 eV — an energy that blocks the wave-to-particle transition because it exceeds the 0.925 eV threshold value of the tau neutrino.

Measuring a particle with an infrared sensor, at photon energy $h\nu$ below 0.925 eV, does not produce full temporal ionization and does not generate a muon neutrino unless the energy exceeds the threshold by at least 0.018 eV.

4.3 13. Wheeler's Delayed-Choice Experiment

The phenomenon above is illustrated by Wheeler's famous delayed-choice gedanken experiment. It was realized experimentally in a configuration equivalent to a single photon, using a single ultra-cold metastable helium atom in a Mach–Zehnder interferometer — an atomic analogue of Wheeler's original proposal (Truscott, Manning, Khakimov & Dall).

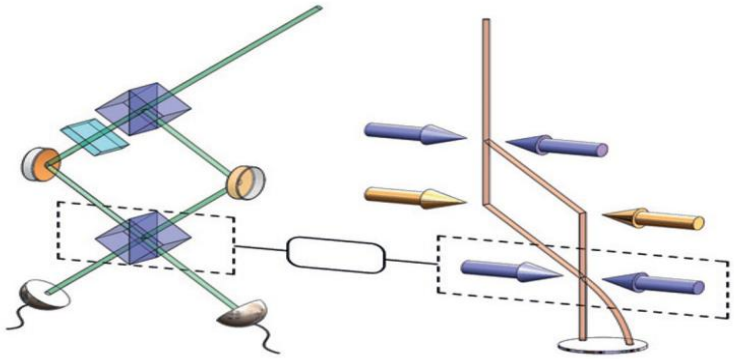


Fig. 1 — Schematic of Wheeler's delayed-choice experiment: the optical Mach–Zehnder interferometer (left) and its atomic analogue (right), after Manning, Khakimov, Dall & Truscott, Nature Physics (local reference nphys3343.pdf).

Truscott's experiment (2015). A single helium atom is sent into an interferometer, and the decision whether to insert the second beam splitter — i.e. whether interference is allowed — is made **after** the atom has already entered the interferometer.

4.4 14. The Transactional Interpretation of Quantum Mechanics (TIQM)

The essence of the Transactional Interpretation, as formulated by Cramer, is that a quantum process (e.g. the emission and absorption of a photon) occurs through an exchange of waves between an emitter and a receiver:

- The **retarded wave** (ordinary wave) is sent forward in time by the source.
- The **advanced wave** is sent backward in time by the absorber.
- Together the two waves form a **transaction**, an atemporal agreement between cause and effect.

Following Cramer's original construction, with one explicit adjustment adopted here: the advanced wave, sent backward in time, is a **longitudinal wave** — the entanglement particle — and therefore propagates at $V = c \times 10^9$, not at an unspecified rate "much greater than c".

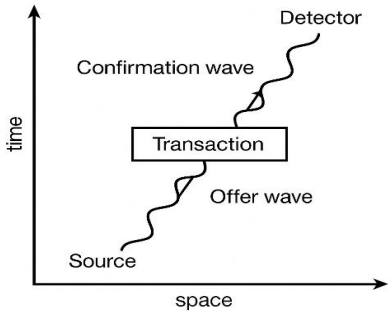


Fig. 2a — Spacetime diagram of the transaction: the retarded offer wave (source to transaction) and the advanced confirmation wave (transaction to detector).

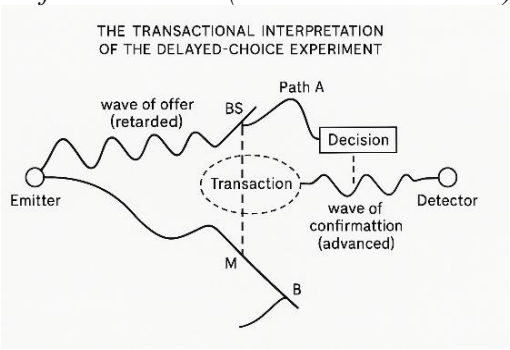


Fig. 2b — The Transactional Interpretation applied to the delayed-choice experiment: the offer wave (retarded) reaches the beam splitter (BS); the confirmation wave (advanced) returns from the detector after the decision is recorded, closing the transaction between emitter and detector via the mirror (M) / splitter (B) paths.

Born's rule: $P = |\psi(r,t)|^2 = \psi(r,t) \cdot \psi^*(r,t)$ ⇒ Transaction $\sim \psi \cdot \psi^*$

4.4.1 14.1 The Transactional Framework

- **Offer wave** ψ_O: propagates forward in time, from the emitter to all possible detectors, at speed c.
- **Confirmation wave** ψ_C: propagates backward in time, from a detector that could register the particle — sent only if absorption is possible — at speed $V = c \times 10^9$.
- **Transaction**: a standing wave formed between the offer wave and the confirmation wave, which determines where the particle actually manifests.

4.4.2 14.2 Hypothesis of Superluminal Propagation of the Advanced Wave

- The confirmation wave travels, at $V = c \times 10^9$, effectively instantaneously back to the emitter on laboratory timescales.
- This removes any measurable delay in the emitter's "knowledge" of the future boundary conditions set at the detector.

4.5 15. The Mathematical Model

4.5.1 15.1 Offer Wave from the Emitter

The atomic wavefunction splits between the two interferometer paths: $\psi_O = \frac{1}{\sqrt{2}}[\psi_A(x,t) + \psi_B(x,t)]$

Each component accumulates a phase set by the potentials $V_A(x)$, $V_B(x)$ along its own path:

$$\psi_A(x,t) = e^{i(\varphi_A - \omega t)}, \quad \varphi_A = \int_0^L k_A(x)\,dx, \quad k_A(x) = \frac{\sqrt{2m(E - V_A(x))}}{\hbar}$$

$$\psi_B(x,t) = e^{i(\varphi_B - \omega t)}, \quad \varphi_B = \int_0^L k_B(x)\,dx$$

4.5.2 15.2 Confirmation Wave from the Detector

Depending on the presence or absence of the second beam splitter, the detector sends back:

- a **coherent superposition**, which enables interference;
- a **single-path response**, which disables interference. $\psi_C = \frac{1}{\sqrt{2}}\left[e^{-i\varphi_A} + e^{-i\varphi_B}\right]$ (interference enabled)

$$\psi_C = e^{-i\varphi_A} \quad \text{(interference disabled)}$$

This wave propagates back at $V = c \times 10^9$, letting the emitter "learn" the experimental configuration effectively instantaneously, on a timescale much shorter than the light-crossing time of the apparatus.

4.5.3 15.3 The Transaction

$$T = \psi_O \cdot \psi_C = \begin{cases} -\frac{1}{2}\left[1 + e^{i(\varphi_A - \varphi_B)}\right] & \text{(interference possible)} \\ -\frac{1}{\sqrt{2}} & \text{(no interference)} \end{cases}$$

Interference condition. The phase difference $\Delta\varphi = \varphi_A - \varphi_B$ determines constructive or destructive interference. **No interference.** The transaction amplitude becomes independent of the relative phase; the two paths are fully distinguishable.

4.6 16. Physical Consequences

- The emitter receives an effectively instantaneous response, via the advanced confirmation wave, about whether interference is allowed.
- Although the choice is made after emission, the transaction only forms if there is consistent phase matching between the offer wave and the confirmation wave.
- Retrocausality, in the ordinary sense, is avoided by treating the whole process as an atemporal negotiation among all boundary conditions, rather than a causally-ordered sequence of events.

4.7 17. Key Observation

The particle's "decision" to behave as a wave or as a particle is not made *at* the beam splitter or at the detector, but non-locally and atemporally, through the matching of the offer wave with the confirmation wave across the entire space-time configuration of the experiment — including settings fixed only in the future.

Superluminal transactional laws. All particles communicate bidirectionally in time through an otherwise undetectable channel, at $V = c \times 10^9$ for the confirmation/entanglement sector — never for the directly observable photon, which remains bound to c.

Partial atemporal determinism. Past and future co-determine each other; events "settle" according to the global coherence of the entire boundary-value problem, not solely through strictly forward-in-time propagation.

Active non-locality. Distant events can influence a local transaction through this invisible channel.

Locally reversible causality. Effects may appear to precede their apparent cause, but only insofar as this remains consistent within a complete transaction.

4.8 18. Summary of Numerical Values (Wheeler / TIQM Section)

Quantity	Symbol	Value
Speed of light (offer/retarded wave, observable photon)	c	2.998×10^8 m/s
Entanglement particle speed (confirmation/advanced wave)	V	$c \times 10^9 = 2.998 \times 10^{17}$ m/s
Tau neutrino energy threshold	—	0.925 eV
Additional threshold for muon neutrino generation	—	+0.018 eV
Optical collapse window	—	400–700 nm (2–4 eV)

This regenerated version fixes the value at the single canonical TauTimp figure, $V = c \times 10^9 \approx 2.998 \times 10^{17}$ m/s for the entanglement particle speed. (already adopted in TauTimp_Gamma_Wave2.md for the gamma field), so that all documents in this corpus refer to the same numerical constant. The speed of light c remains unchanged for any directly observable wave (the offer wave, i.e. the photon or physical atom); only the advanced/confirmation-wave sector — which carries no independently readable message, per the no-signaling requirement discussed in Rezumat_Pettini_Entanglement_RO.md — receives the superluminal value.

4.9 References

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